

Stellar-Chill Electrocaloric Refrigerator

Cost Analysis per Unit — CSMFAB 004

Category: Aegis Home — Kitchen **Unit Basis:** 1 standard under-counter unit (120L) **Date:** June 2026
| **Document Type:** CSMFAB Cost Study V2.0

1. Product Summary

This cost analysis is derived from the V2.0 specification and v1to2 versioning document for this product. All material costs reflect 2026 market pricing. Three production scales are costed: - **Prototype:** 1-5 units (single-setup, high labor, no tooling amortization) - **Small Batch:** 100-500 units/year (semi-tooled, moderate labor) - **Production Volume:** 5,000-50,000 units/year (fully tooled, optimized labor)

2. Bill of Materials (BOM)

Component	Quantity/Unit	Unit Price	Extended Cost
Aegis-C BFRP monocoque casing (Elium®, 3mm)	8 kg	\$25.00/kg	\$200.00
PVDF-TrFE-CFE electrocaloric film stack (5 layers)	0.15 kg	\$800.00/kg	\$120.00
Vacuum Insulated Panel (VIP) module set (6 panels)	6 kg	\$45.00/kg	\$270.00
LiFePO4 thermal buffer battery (0.5 kWh)	4 kg	\$100.00/kg	\$400.00
PEEK CF40 shelving unit	1.2 kg	\$280.00/kg	\$336.00
MXene EMI shield (interior)	0.03 kg	\$2000.00/kg	\$60.00
YInMn Blue exterior glaze panel	1.5 kg	\$75.00/kg	\$112.50
KNbO3-BaTiO3 temperature control transducers (6×)	0.02 kg	\$180.00/kg	\$3.60
PMMA optical fiber sensor loom	0.3 kg	\$25.00/kg	\$7.50
Compressor (none — solid state only): \$0	—	\$0.00/kg	\$0.00
BFRP refrigerant-free thermal circuit	1.0 kg	\$25.00/kg	\$25.00
BOM Subtotal			\$1,534.60

3. Labor, Process & Overhead Costs

Cost Element	Prototype	Small Batch	Production Volume
BOM Materials	\$1,534.60	\$1,258.37	\$997.49
Direct Labor	\$520.00	\$380.00	\$220.00
Overhead (15% of labor)	\$78.00	\$57.00	\$33.00
Quality Control & Testing	\$62.40	\$30.40	\$11.00
Packaging & Logistics	\$46.04	\$38.37	\$30.69

Tooling & Equipment Notes: VARTM mold for casing: \$2,800 NRE. EC film lamination: \$80/unit.

4. Per-Unit Cost Summary

Scale	Total COGS	Suggested MSRP Range	Gross Margin
Prototype (1-5 units)	\$2,980	\$4,768 – \$6,556	35-55%
Small Batch (100-500/yr)	\$2,100	\$3,150 – \$4,200	33-50%
Production Volume (5K-50K/yr)	\$1,580	\$2,212 – \$2,844	28-45%

Market Reference: \$2,500-3,800 (premium solid-state market; no HFC, no compressor failure)

5. Key Cost Drivers (V2.0)

Driver	Impact	Mitigation
ZrB2/Si3N4 UHTC ceramics	HIGH — specialty powder \$350+/kg	Bulk procurement; domestic synthesis pilot (Japan NIMS CO2 route)
MXene Ti3C2Tx	VERY HIGH — \$2,000/kg at 2026 scale	Tile pattern minimizes usage; price projected -60% by 2028 as production scales
Elium® thermoplastic BFRP	MEDIUM — recyclability adds value; \$12/kg resin	Phoenix Protocol recycling recovers 100% monomer; net cost offset
Tooling NRE	HIGH for first production run	Paid by pre-orders or grant funding; amortized quickly
Flash Sintering process	MEDIUM — shares SPS equipment	Regional ceramic sintering facility sharing model

6. Phoenix Protocol Circular Economy Value

At end-of-life, material recovery adds back-value: - **Recovered material value per unit:** \$276 – \$537 - **Recovery reduces net lifecycle cost by:** 15-30% - **Critical minerals (In, Y, Co, Ti) recovered** via ionic liquid extraction